Apply AI Project Proposal

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Topic: COVID 19 Vaccination Rates in the U.S

Research Question: When will 90% of the U.S population be fully vaccinated?

Sources of Bias:

- People are unable to get the vaccine (e.g. those who may face severe allergic reactions/medical conditions, location to provider)
- Potential issues with processing the data (margin of error, ie: incomplete/missing data for certain days)
- Homelessness (some states' distribution plans do not include the homeless population, "silent on vaccinating homeless" - Professor Hee Soun Jang, UNT associate professor)

NOTE: We are focusing on the population that is eligible to receive the vaccine. Children under 12 are not included in the dataset and are not considered for our prediction.

Summary:

As COVID-19 continues to be a prevalent global health issue, affecting populations around the world, we hope to see the direction in which our society is going in regards to combating the virus. With vaccination plans continuing to be developed and adapted, we find this topic to be very relevant. Our group chose Gabriel Preda's "COVID-19 World Vaccination Progress" dataset on Kaggle (Gabriel Preda, COVID-19 World Vaccination Progress, 23 October, 2021, Kaggle, https://www.kaggle.com/gpreda/covid-world-vaccination-progress), which is frequently updated with current numbers on total vaccinations.

This dataset is sourced from the organization "Our World in Data", which gets the official metrics from governments and health ministries worldwide, including the United Nations World Population Prospects for per-capita population estimates. While it does contain data from multiple different countries, we hope to focus and extract the data specifically tracking the progress within the United States.

Machine Learning Algorithm to be used: Regression Modeling Techniques, Polynomial and Nonlinear Regression Models

Datasets:

- Kaggle: COVID-19 World Vaccination Progress
  - Our World in Data
  - Github

Sources:

- Towards Data Science: Understanding Regression Using COVID-19 Dataset
- Towards Data Science: A Simple Guide to Linear Regression Using Python
- Towards Data Science: A Complete Guide to Linear Regression Analysis
- Yale: Linear Regression
- Pew: States Fail to Prioritize Homeless People For Vaccines
- USA Facts: COVID Vaccine Progress Tracker - Vaccinations by State
- NY Times: Interactive World COVID Vaccination Tracker
- Statista: COVID-19 Cases Worldwide, Recoveries, Deaths
- Statista: COVID-19 Total Number of Cases and Deaths within the U.S
- Research Letter (Dr. Sacarny, Dr. Daw): Inequities in COVID-19 Vaccination Rates within the U.S
- The Washington Post: Why the coronavirus vaccine may not be accessible for the people who need it most